ActInf Textbook Group ~ Cohort 3 ~ Meeting 15 (Chapter 6, part 2)

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All right, greetings, everyone.

It's June 20th, 2023.

We're in our second discussion on Chapter 6.

So we'll first just have any general comments or anything.

Then we'll turn to the questions table and look at what questions we didn't get to last time and just kind of revisit, maybe condense some questions or just see what we can do with what is here.

So anyone just want to make any general comments on Chapter 6?

I believe this is the second session on Chapter 6, right?

Correct.

Or maybe I must.

Okay.

No, no.

Yeah, it is.

Yeah.

There was only one onboarding needed for the welcome back here.

So we started one week faster into the rhythm.

Okay.

So it looks like these were the questions.

I remember we left off with this one.

Let's just develop this into what was into a question and then we'll continue on.

Okay.

Is there a common or good representation or rubric for evaluating generative model, generative process?

Let's just change it to generative models and generative processes.

Anyone want to give a first thought or just some other related question?

Okay.

Yeah.

Well, go on, please.

Sorry.

No, that's okay.

That's okay.

Yeah.

I don't know.

I'm not sure about what's the exact criteria for evaluating generative models or generative

processes.
But one thing is for sure that, well, I mean, one of the main, at least when evaluating generative model, one of our main criteria should be how closely it tracks our situation of interest and specifically how relevant.
And specifically how relevant it is for addressing the question we're trying to, I mean, we're trying to examine.
So it's not, I don't think it's something clear cut or, I don't know, written in stone.
And based on the context and the situation, the evaluation criteria rubric should be different, I suppose, both for generative model and generative process.
Yeah.
So again, we'll revisit this generative model generative process question later following the recent live stream.
But generative process, working with generative process as the underlying process that gives rise to the observations, this would seem to be more adequate to the extent that it can better describe the measurements.
Generative model similarly is being evaluated based upon its ability to fit to the generative process as well as phenomena of interest.
So not just to fit visual data, but maybe to model something like some illusion in visual.
Then there's a wide range of more general statistical modeling techniques.
So first is like the ultimate grab bag, which is just how relevant is the overall modeling.
Then there are statistical evaluations of model adequacy and model selection.
IQ key information criterion, Bayesian information criterion, Bayes factor, hierarchical likelihood ratio test.
If you have a parametric model, bootstrap and non-parametric statistics, just general modeling.
And then taking that more out of the statistics into like the engineering model lifecycle, then that's where you can think about validation, verification, validity, quality, et cetera.
Systems engineering model lifecycle.
Any other thoughts or questions on this?
Yeah.

Okay.
So I'm coming from kind of an early career data scientist background here, but I mean,
as far as I mean, what I find attractive about active inference and doing things as generative
models is the attempt to, you know, describe or explain what's going on in the data.
And that being said, you know, in terms of a lot of many applications of like data science,
machine learning, the emphasis is upon prediction.
And so are these listed statistical metrics you have here, base factor, BIC, et cetera?
I mean, are those basically like your error measurements?
Are they your, how to say, just attempted at measuring like accuracy of prediction or,
or is there some trade-off between explainability and prediction?
Maybe that makes sense.
Yeah.
Yeah.
Just to kind of summarize these.
So when you have nested parametric models, then you can evaluate whether two models are
better in terms of including or not some parametric factor and then testing for their likelihood
ratios.
So this is a pretty straightforward test, but it's pretty limited to just strictly nested
parametric models.
So just, just getting that out of the way in general, we're in the Bayesian setting.
And so the, one of the advantages of the Bayesian setting is that you can compare two different
models.
That have totally different architectures on the same data set.
So one way that that's done is with this IKKI information criterion or the Bayesian information
criterion.
Both of them do something very similar, which is they reward model fit and they penalize
the number of parameters.
And so actually in that way, it's a lot like equation 2.5.
You want to reward the fit, but penalize the complexity of the model.
Base factor is kind of like the HLRT, but it doesn't have to be ratios of nested models.
You can just compare the relative evidence in favor of one model or another.
So this is used in structure modeling and, and structure learning.
Then in terms of, but this is all like kind of like testing amongst a portfolio of models,

which ones are better.

Then how do you really get into the space of like, well, how effective is this model?

You could test on a new data set.

You could do all the regular techniques like cross validation, leave one out, et cetera,

or test on a new data set.

But then the very interesting question is how do you establish the efficacy of an action model?

Because, you know, it's, it's not just a passive inference model.

So there you would need to get into like benchmarks, like the OpenAI gym and other,

other settings where you can actually test the efficacy of an action model against some test input,

rather than just a recognition model and some input.

But that gets so situational that there aren't as, as there's not like general,

not the, there's not as many general considerations there.

Another strategy is actually to, even if it seems like a passive inference problem,

like the MNIST digit data set, you could frame that in terms of the, the labeling is an action.

So sometimes you can then take what seems to be a passive inference problem and then frame it as an active inference problem.

So then you don't need to worry about these kind of like out of sample, novel context action models.

You can just kind of use standard benchmarks.

And also, as far as I know, probably the only serious work on benchmarking the performance of active inference models

are the work of Theo Fischampion and his colleagues, branching time active inference, which is,

and his colleagues branching time active inference, which is, they've proposed several different variants of branching time active inference.

each of which I mean with increasing performance in terms of their benchmarks and aside from that

I'm not aware of any other work that takes this benchmarking study seriously enough to

to be reliable I guess yeah surely there's a lot of proprietary work in this area

the paper discussed in in live stream eight scaling active inference so here they they use

like the kind of several standard testing environments I forget what the pendulum

the hopper and then I think like the mountain car or maybe maybe that's not this one but

they test the standard um AI tests

all right we're just looking just you know going through the ones all right are the transitions be

independent from the emissions so for um uh context we're talking about figure 7.3 discrete

time active inference are the transitions independent from the emissions and what does the next observation depend on

short answer yes they're independent that's the sparsity of the base graph so we can clarify give

a cleaner textual answer but yes the conditional independencies which is the sparsity of the base

graph which allows us to do factorized variational inference and get all of these advantages

that's exactly what we're looking at so this visual graphical model is the sparsity architecture of the conditional um

independencies

so yes

a and b are conditionally independent

conditionally independent based on what hidden state

that's how markov blanket works

conditionally independent on the blanket a and b are independent

that's true of all bayesian graphs

so without worrying about the markov blanket and the interface and the cybernetic agent and all of that

just

any node that intervenes between two other nodes is the markov blanket in that setting

and then some other node you know something else is blanketing it from something else

but yes they're conditionally independent what does the next observation depend on

well a matrix is the emission matrix tail of two densities it can emit from a hidden state or it can

recognize from an observation so any given observation is only

dependent upon the hidden state at that time and the hidden state at the next time

is only dependent upon the transition matrix the transition matrix is uh has a slice for every action that can be taken

every policy makes a slice in the b tensor so it's like temperature in the room hidden state

thermometer observation

turn on the heater or not and there's some b matrix for the heater is on and there's some

b matrix for the heater is not on those are two slices in the same object and then policy selection means

which matrix which sub matrix would be which slice of b should we multiply this to get to the next time step

and then what observation would i expect there

okay good good um kind of standard question okay what is the relevance of thinking about good regulator theorem

for thinking about the generative model so from cybernetics good regulator theorem originally stated

every good regulator of a system must be a model of that system or more accurately every good regulator must contain a model of that system

here's a few quotes from a textbook

one way to approach this again these are all kind of big questions but we can just see them many times

what happens to bad regulators well information processing sense making decision making has a non-zero informational cost

land hour limit there's a certain amount of actual jewels it takes to write and erase

a bit it from bit chris fields everything that is in that area so information processing is never free

so in a dissipative and even adversarial universe

to fail to regulate

is to fail to exist

how do we operationalize regulation in this setting

that's going to be revisiting some non-equilibrium steady state

so if we're

an observer looking at a system

in order for us to measure it as diff as signal relative to noise

it has to be persistently

remeasurable

over the background so we have to repeatedly measure it

or an organism let's just say

can remeasure itself

so minimizing surprise about its homeostatic temperature

is being adaptive

now if temperature were just flat

and so you could be unsurprised at homeostasis by doing nothing

you'd get

lazy agents

but if temperature was really variable and contextual and there was kind of these non-linear cues in the environment and all of that

then in order to reduce surprise

about temperature

you'd end up

coming to effectively

have a model of the causal nexus that gives rise to observations

so the structure of the generative model doesn't have to be the structure of the generative process

however they may come to have certain isomorphisms with each other

um or at least statistical regularities

if there's actually a 24-hour cycle in temperature

then you're going to see some kind of oscillatory model in the generative model

so

there's more to um

say there but oh and this is one good note perhaps rather than good or bad regulator the uh language of morality

or preference

the language should be accurate and inaccurate

in effective and effective viable inviable skillful unskillful so

there's always many ways to see it but basically

if the generative model

in the in the limit case it just totally knows the causal architecture of the world

that's the easiest way to be absolutely unsurprised
and to have things as you expect slash prefer
is literally know how they're going to play out
that's not plausible
we use coarse graining and approximations and heuristics
like variational inference
so the best we can do is just iteratively optimize
towards
bounding surprisal
so it's kind of like an empirical
optimizable heuristic for being a good regulator
without getting too bogged down into like the philosophy and the exactitude
the point is the ones that do well enough to live live
ones that don't do well enough don't
but active inference is in the lineage of cybernetics
so it's unsurprising that good regulator theorem requisite diversity
viable systems models
a lot of things in the cybernetics ontology have a natural home in the active ontology
that they're you know they're talking about the same territory adaptive agents so it's not
surprising that like they don't invalidate each other or anything like that
and actually i think uh it's one of the reasons cybernetics
has experienced kind of revitalization in recent years especially in the past couple of years so
yeah cool
all right how can organs other than the brain be making inferences
so there's a few angles on this the first um ang or ali or anyone else want to give a thought
uh one thing to point uh is the sense of kind of semi-pan computationalism uh that's been um
uh if not pan computationalism but something that uh bestows um inference not only to complex
self-organizing systems such as the brain but even uh to very simple linear systems even
uh system as simple as uh an inert rock uh so basically it's it it covers a spectrum uh there isn't i mean uh
from the point of view of fep there isn't anything that's specifically unique about the brain uh in other
words the same mathematical technology can be applied uh both to inert rocks as well as the uh the brain uh so
in my opinion the uh this question should be turned on its head and it's not that
uh how does inference can be seen in other uh systems that are simpler systems but rather
it should be i mean the relevant question should be uh how does active inference or
the notion of inference inactive inference literature applies to
uh all of those situations uh so one way to do that is to define a kind of uh sparse coupling between
the environment and the agent and uh define variational density as the internal states of the system

and also obviously the external states of the uh environment would be the states that needs to be
tracked by those variational densities and uh in this case there isn't anything inherently different
between the way the brains uh or i don't know even sentient agents uh
uh somehow undertake uh inference uh from the way that uh the inert rocks can partition uh their states into
internal and external states but the main difference would be the way the markov blanket in those simpler
systems can act as uh as a kind of statistical boundary that allows for non-causal relationships i'm sorry
non-linear uh causal relationships uh as opposed to as opposed sorry linear causal relationships as opposed to
non-linear causal relationships as observed in uh complex agents or systems uh so
yeah that's uh some of the main distinctions between those awesome yeah and then the classic paper that we
we often return to uh like ali mentioned the inner rock so this paper has the uh kind of visual taxonomy
from simpler to more sophisticated agents
that's fine
uh
sometimes the papers are in white sometimes they're in black
here
inert systems active classical conservative systems strange systems with world models of their own and all
of this um just but but this is a great response which is in a pan pan computationalist or pan
cognitivist pan inferentialist world then how does active inference apply one other
angle is like let's just think about the liver or the pancreas and blood sugar that's our system of interest
realism is like is the pancreas actually doing inference on blood sugar and people can have a range of
opinions but in a pan cognitivist world the answer is yes or one can kind of pull back and just say we're
gonna model the pancreas as doing inference on blood sugar so it's it is no different it's just an interpretation of the same statistical apparatus
basically
uh but there might be a setting where um scientific realism is is more justified there's multiple lines of converging evidence um the model is is uh comprehensive
and so uh and being used to generate unique explanations and predictions versus like we're just going to do a linear relationship between these two things in the public health
and so then we're not going to confuse the linear model with
those with the actual causal architecture but there are situations that you can design or analyze
where the sparsity structure of the system which as la points to is what grants it all these interesting properties
it is becoming known to an extent that within
two or three or four sigma it's like we're starting to talk about how it is not mistaking the map for the
territory that's the map territory fallacy but also not doing some kind of map denialism
which is the map territory fallacy fallacy
and that's the paper which i'll add into by maxwell et al

and also if i may add another point uh with regard to the distinction between realism and instrumentalism
uh i believe uh at least uh in the realism school uh probably the most relevant or well-situated
stance to uh reframe active inference uh on the i mean uh reframe its ontological status is structural realism
as argued by uh uh benny and others in several papers nice yes he's joined several discussions and it's it's great to
um
okay so there's some quotes from the book but yes basically
um forget organs other than the brain how could anything be doing inference and every
and and then another angle to take is just like the way that a baseball clock uh computes a parabola
like you don't need to think that it's doing it like a calculator
but it but that's kind of like naturalizing the computation that's a path of least action in a spatial
space but also you could have a path of least action in a cognitive space that's bayesian mechanics
and then you could have more realist or more instrumentalist angle yeah yeah please
uh sorry just uh one one thing that uh somehow uh i mean uh gets confused is uh some sometimes people
think feps claim is that okay so if um a self-organizing uh i mean uh if fep formalism applies to
a self-organizing a system then it should persist through time uh by preserving uh its markup blanket
intact but actually uh the claim of the fep is the other way around so it's a main assertion is that
if anything persists through time how does fep applies to it so in other sense uh fep doesn't
provide any justification per se for describing the way that the system persists through time but
uh we take the persistence of the system through time as the premise and then apply fep to somehow
investigate the implications of that premise through bayesian mechanics and fep formalism
yeah without the implicit or explicit acknowledgement of the persistent existence of what you're measuring
or yourself you don't even have something to discuss and a lot of times when people just
start to specify what they mean by like the nouns and the verbs and the adjectives they use
that's a deflationary approach what if there's something that comes into being so so fast in
between moments and it's not being measured and you can't detect it it's like well then how do you know
it's there that's not within that statistical sensing apparatus so you have to take the existence of the of the thing
to be
to be primary in modeling something
and actual constant has a lot of work in that area
uh daniel could you please pull up the path integral paper on page uh four if i may
uh
and it's like well if we're talking about something that exists even in a mental geometry
that's what we want we want to start with if it exists even hypothetically
so people say can it really describe all things it's like well all the things that exist
or could exist we could use this modeling approach to model
and talk about what properties it must possess but if you're willing to open the door to things

that don't or can't exist then of course this guy's the limit on what properties they do or
don't possess because they might it's like the you know non-elephant animals things might not
exist for a huge variety of reasons however things that exist either from an external
measurement or from self-measurement they must be acting as if they're self-evidencing
reducing surprise relative to a generative model path of least action through an information
geometric space if it isn't doing that you're not going to observe it and that's where we get the
particular partition and the particular partition in the in the particular physics
slash bayesian mechanics since carl fristen's 2019 monograph a free energy principle for a particular
physics that was like a big inflection point in this line of research whereas in the 2016 to 2018
there's a lot of qualitative work and the philosophical work applying to multi-scale and nested systems and
self-organization amidst other empirical simulations happening continually in all of this
so reading reading works of different times like different pieces are kind of going to be made clear or not
what is the role of precision in nested models
well here all uh and also by way of demonstrating the the transferability of active inference generative models
these three figures are discussed in live stream 28 from the original paper smith et al computational
phenomenology so this is a static perceptual bayesian model we have hidden state observations and we
here as a modulator literally a neuromodulator on the a matrix tale of two densities recognition matrix
generative model you have the precision so um precision is one over the temperature so whether you think of
it as like higher temperature is lower precision or lower you know or vice versa they're they're they're the same
when your precision is low your temperature is high a gets blurred out the high temperature limit is
like a is becomes flattened whereas in the low temperature limit a becomes sharpened and so this is a common method in uh recognition modeling okay
now they move into adding action in so here there's the policy figure 4.3 figure 7.3 everything we've looked at
and then they move to this um nested model now in the sandved smith paper it was being mobilized in terms of
basically phenomenology meditative experiences direct sensory perception you know visual perception and eye
and eye cacades attention and eye cacades attention unreportable attention and then awareness at a third level observing that
and so the you you can add an uncertainty um like a precision on any variable you could have an uncertainty on your prior d
you could have uncertainty on all kinds of things but sometimes uncertainties on specific variables
become associated with like certain cognitive phenomena in this case they're using it to describe
introspective and explainable ai systems taking the exact same figures exact same formalisms just
moving them into the ai setting so um precision on a and here's that live stream 28 and the slides and everything
precision on a is like sensory ambiguity um precision on g so how how um precise are you on the the free energy

um that is associated with the affect in this uh sophisticated affect here's some discussion on embodiment and the
alexander technique um so suffice to say precision nested modeling is basically like the general precision
concept which is basically used everywhere this is how we fit bayesian models we have some hyper prior prior
distribution on a parameter and if that prior distribution is too sharp then you're you're basically that's a
it's a pathology of the of the prior it's too precise if it was too imprecise you have an underfit model so
it's like that and especially in nested models precision plays a role in kind of gating just in this case it's a
general question so you can make it do anything basically but here the precision on a is like gating
let's just say the first and the third level so if there's no attention being paid to sensory input
don't be surprised that they don't come up to attention but if the if this um gating can be like
super direct and forceful then you'll have propagation of informational causality on this network up to the higher level
page 113 of the textbook
all right the decision so this is about planning as inference which one of these questions um was also about
the decision whether to model alternative futures counterfactual futures contingent upon policy
selection how would the world change if i did this or that is largely tied up with the choice between
between discrete and continuous models because the idea of selecting between alternative futures defined
by sequences of actions is more simply articulated using discrete time models all right so
figure 4.3
figure 4.3 is like the rosetta stone it juxtaposes the discrete time continuous discrete time generative
model and the continuous time generative model so i'll just copy this in
in the discrete time model
you have s t minus one s of t s of t plus one so
futures and pasts if it's sophisticated active inference are being explicitly modeled
so it's like
what would happen at seven o'clock if i did this
that is explicitly addressable with a discrete time model of the correct time horizon
so you have to explicitly say i'm talking about an hourly model and seven depth and so then you can
branching time active inference just like a chess algorithm it's dealing with like the branching
structure because if you have a lot of affordances
then then you have and you have a lot of time depth you can imagine this this is a combinatorial explosion
um in contrast
though these two generative models are shown here to emphasize their structural similarity
the continuous time generative model is actually not explicitly modeling past present future time
steps rather it's modeling the generalized coordinates of motion
x hidden state x prime rate of change x double prime second derivative so in that way it's a lot
more like a taylor series expansion so a taylor series expansion technically even a taylor series

expansion of you know depth one it has an answer for every single point in the number line but that
doesn't mean it's a good one um and so the continuous time models kind of trivially have something to say
about all possible time steps just by analytic continuation of the taylor series expansion in the generalized coordinates of motion way
however you don't necessarily like know how accurate it is for any given point
it also doesn't have explicit counterfactuals
so if you do some taylor series expansion it says yeah at x equals three over y is five
but then if you say well what if it were different the only way you could do that would be
change the parameters of the taylor series and recalculate it at that point
so the model itself is not exactly doing these like what would happen if i did that
so futures are explicit in discrete time models and interpretable explicitly whereas in continuous
time models pasts and futures are kind of trivially predicted such that it doesn't really make sense
to talk about just um counterfactuals in the same exact way
and these models can be hybridized and and fused
which is in um chapter eight
can we generate a coded template looks like yes but there's of course more information
but it's an important question
how is temporal depth specific to planning does temporal depth in perception make any sense
what is temporal depth so temporal depth is how many time steps are the models considered in the
discrete time case um does temporal depth make sense in perception well
perception is always modeled as instantaneous however depending on the temporal scale of
of a model that perception may be instantaneous over a certain temporal thickness
power why is temporal depth specific to planning
so in planning as inference policy selection as inference you're selecting or sampling from policies
based upon their expected free energy
in order to plan over a given time horizon and actually evaluate
the um the playouts you need to have a temporal depth of that amount
thomas parr in his um bookstream
so early implementations were coming from the filtering of a generalized filtering approach
maybe it wasn't generalized at that point but particle filtering approach and then equipped
these models with action then people started thinking about how to get sequential dynamics
predator prey lock of ulterra models winterless competition neural darwinism in these situations
there's an emergence of sequential recurrent continuous dynamics like a limit cycle of more than two
then people wanted to model explicit long-term planning and so from around you know
20 uh 14 or something there was a lot more development in the discrete state space model which enabled the
well understood partially observable markov decision process
and a lot of the characterization of planning as inference explore exploit

um

then

later developments supported

um nested models here there's a discrete time on top and a discrete time on the bottom but also it could be

um continuous time on the bottom

um oh yes then continuous state models were reintroduced

as the lower levels of higher level discrete time models

and um that is kind of like the folk psychology live stream 46 active inference does not contradict folk

psychology discrete time decision making up top

you know discrete time and discrete decisions

and then more in the sensory motor it's more like continuous time continuous action

okay

new question

figure 6.1

all right

particular partition

how is the mutual interaction between active states and sensory states meant in 6.1 so the

bi-directional line here

can they mutually change their states without impacting neither internal nor external states

yes they could you could have um nodes that are these nodes are in communication with each other

in general this image it's more important what it what it doesn't show so there's no backwards arrow from

external to active or from internal to sensory you can't have your thumb on the scale

and the symmetry of whatever that is from the outside and the only other constraint is

no telepathy no telekinesis the only way to receive information about external states is through sensory states the only way to act on

empathize from empathize

empathize

empathize

!

empathize

same importance. So if you want to design a computer system where the sensory states comes in one computer, and this is a second computer, and this is the third computer, so there's no direct edge between sensory and active states, you can create that causal architecture.
What is it supposed to model? Nothing. Modeling of supposed things is done in chapter six and beyond, but in general, this is not trying to model any specific situation despite the brain in the world. Could they circle, mutually modifying their states, and then eventually arrive at a state where they change the external internal states? Yeah, maybe they have a faster timescale or something, or maybe they're in communication with each other, and then sensory states ends up influencing internal states via active influencing id, and so yes, it could happen. Would that mean that we can model with this trick any arbitrary behavior? It's not necessarily a trick, it's just a particular partition. Would that allow us to model Turing equivalence? I'm not sure if there's an understanding of what formally demonstrates Turing equivalence, but I believe it's possible, I think, as a Turing architecture, as far as I understand it, abstracted from the von Neumann architecture, is basically just saying, like, you can do a Turing tape, so I don't see why not. Why would we need that?
512k ought to be enough for anyone, right? I don't know why we need it. We expect it.
But, yeah, these graphs in general are more like the space of the possible, with the important caveats that were mentioned with the no thumb on the scale, and the mirror, and the no telepathy, no telekinesis. But how relevant these edges are, or what systems have what behavior, or what cognitive phenomena are granted by what dynamics on graphs, there's just no general answer to those things.
Because if you do a graph for one setting, it's going to be different.
What did you mean when you said that any entree could be ordered with any side dish?
Like that, in this recipe theme that we're in.
Models that include continuous or discrete variables are both.
What is meant by variable, states, or observations?
How can states be continuous?
As those here are probably familiar with,
states or observations could be, like, discrete,
like 0 or 1,
or any integer,
or it could be a continuous number.
Computers discretize continuous functions to approximate them.
But there's so many exciting directions with unconventional computing,
analog computing,
MEM computing,

et cetera,
that, like,
there's more to it, but just simply,
it could be any type of variable.
Okay.
Okay.
This is a new question.
This is,
this is,
let's maybe look at this one in closing.
If the external state is unknowable,
how can we set up a generative process?
So that's what's generating the observations.
When our experience is based solely on
the process of producing variational free energy
based on predictions and sensory inputs across the Markov blanket.
This person has just described the challenge of life.
I don't think that there is a specific answer to that.
I think this is literally a restatement of the particular partition.
If this was just said,
the challenge is based,
is to given the unknowability,
direct unknowability of internal states,
the challenge and the opportunity
is to set up a generative model
based solely on the process of reducing free energy
based on predictions and sensory inputs across the blanket
and adaptive action.
I'm not sure if they meant process or model here
because setting up a generative process
is something that the human modeler does
when they're designing a simulation,
but not like what the animal has to do.
What I'm trying to get at here is
we are making an assumption about the generator of sensory input
when making the model.
Yes.
But that model can never be accurate.

Well, it absolutely can be accurate.
It's never going to be a one-to-one map as the territory,
but of course it can be accurate.
We don't need to understand,
have an atomic simulation of the sun
to have a generative process
of how many photons are going to hit my window
tomorrow at 7 a.m.
So there's a core invalidity in the notion
that we set up a generative process model.
Not sure what ontology mixing is happening,
but almost by definition,
we're introducing an error in the model
by setting up the parameters
for the generative process of the generative model.
We remove the fundamental aspect of actin,
which is the dynamical system has only one thing it can do,
and that's to reduce free energy.
So there's some good points and some mixed things.
So what it does is the action it takes in the world.
Variational free energy is just a tractable computational heuristic.
Yes, if you set up the generative process
to be a continuous number between 1 and 10,
and then the generative model to do the exact same
to kind of have pre-structurally learned the problem,
then it's going to be a simple simulation.
Whereas you could have a more sophisticated simulation
that involves structured learning
as part of the agent's generative model.
Are we not introducing information to the model
that would not be available in the real world
if we actually defined the generative process?
Yeah, you can give any model too much information, essentially.
So if the model actually knows what is really happening,
then if you were doing a video game
and you had an agent with supervisory access
to the other players' inferences
or even control their actions,

yeah, that's not going to work in a tournament.
But here with a particular partition,
we can actually know the information encapsulation.
And Majeed Benny explores that
in terms of the partial information encapsulation.
So are we not introducing information to the model
that would not be available?
It's your restaurant.
Do whatever you got to do with the recipe.
If you want a pedagogical example
that just makes some clean graphs
and is very straightforward
and doesn't engage the complexities
of like sophisticated cognitive structure learning,
that model's not going to complain.
If you want to do
strange loop reflexive structure modeling
for ambiguous inputs
of unstructured multimodal data, etc.,
then that is your challenge.
So for many people,
it's just one closing comment.
In many people,
this is the first time they've been exposed
to statistical modeling
and iterative modeling formally.
So that's why there are often questions
that are like less about active inference,
but sometimes crop up
about basically using statistical modeling overall,
which is a great thing
because these are challenging
and areas with a lot of implicit
and tacit knowledge.
So thank you, fellows.
Looking forward to next discussions
and heading into Chapter 7.
Next week, Ali,

maybe we'll do our .0

for Chapter 7 and 8,

but not in a hurry,

but we'll figure it out.

Sure.

Thank you so much.

Thank you.

Thank you.

Bye.

Bye.